

Zhenyao Cui

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Research Interests

My research focuses on transferable and real-time neural decoding. I am especially interested in scaling laws for foundation models of physiological signals, unified models that work across subjects and heterogeneous recording settings, and multimodal learning.

Education

Huazhong University of Science and Technology

Sep 2024 – Jun 2027

M.Eng. in Intelligent Science and Technology (expected)

- GPA **3.89/4.00**; selected for direct graduate admission through the national recommendation scheme.
- Advisor: Prof. **Dongrui Wu**, IEEE Fellow and Editor-in-Chief of *IEEE Transactions on Fuzzy Systems*.

Huazhong University of Science and Technology

Sep 2020 – Jun 2024

B.Eng. in Automation, Elite Engineer Program

- GPA **3.91/4.00**; Outstanding Graduate of HUST.

Publications

[†] equal contribution.

1. **Zhenyao Cui**, Siyang Li, Siyuan Kan, Yudi Wang, Fei Luo, and Dongrui Wu. “A Fuzzy Set Based Classification-to-Regression Extension Framework for Transfer Learning and Its Application to Brain-Computer Interfaces.” *IEEE Transactions on Emerging Topics in Computational Intelligence*, Early Access, 2026.
doi:10.1109/TETCI.2026.3683713
2. Siyang Li[†], Jiayi Ouyang[†], **Zhenyao Cui**[†], Ziwei Wang, Tianwang Jia, Feng Wan, and Dongrui Wu. “Backpropagation-Free Test-Time Adaptation for Lightweight EEG-Based Brain-Computer Interfaces.” *IEEE Journal of Biomedical and Health Informatics*, accepted for publication, 2026.
arXiv:2601.07556
3. Dingkun Liu[†], Yuheng Chen[†], Zhu Chen[†], **Zhenyao Cui**, Yaozhi Wen, Jiayu An, Jingwei Luo, and Dongrui Wu. “EEG-FM-Compass: Progress, Benchmarking, and Future Directions for EEG Foundation Models.” *National Science Review*, nwag466, 2026.
doi:10.1093/nsr/nwag466
4. **Zhenyao Cui**, Siyuan Kan, Siyang Li, Ziwei Wang, and Dongrui Wu. “SCORE: Subject Coordinate Recovery for Label-Free Cross-Subject EEG-to-Image Retrieval.” Preprint; manuscript under review, 2026. arXiv:2608.19134.
5. **Zhenyao Cui**, Siyuan Kan, Dingkun Liu, and Dongrui Wu. “Beyond Trial Averaging: Anchoring Neural and Visual Representations for Few-Repetition Brain-to-Image Retrieval.” Preprint; manuscript under review, 2026. arXiv:2608.19128.
6. **Zhenyao Cui**, Jiahui Chai, Siyuan Kan, Yudi Wang, and Dongrui Wu. “MCL-Align: Marginal-Conditional-Label Alignment for Cross-Subject EEG Regression.” Under review, *IEEE Transactions on Biomedical Engineering*, 2026.
7. Siyuan Kan, Huanyu Wu, **Zhenyao Cui**, Fan Huang, Xiaolong Xu, and Dongrui Wu. “CMCRD: Cross-Modal Contrastive Representation Distillation for Emotion Recognition.” *IEEE Transactions on Affective Computing*, 17(2):1837–1846, 2026.
doi:10.1109/TAFFC.2026.3664341

Research Experience

Transferable EEG Decoding

Apr 2024 – Present

Laboratory of Brain-Computer Interfaces and Machine Learning, HUST

- Proposed **C2R**, a fuzzy-set framework that extends class-conditional transfer learning from classification to continuous EEG regression, and derived a target-domain risk bound based on fuzzy class-conditional probabilities. (*IEEE TETCI, Early Access, 2026*)
- Co-authored **EEG-FM-Compass**; contributed to the literature review and dataset curation, and evaluated the regression performance of multiple EEG foundation models, including **LaBraM** and **CBraMod**, on the SEED-VIG continuous PERCLOS prediction task under cross-subject LOSO and 10% within-subject calibration, comparing full fine-tuning with linear probing using RMSE and correlation. (*National Science Review, 2026*)
- Proposed **MCL-Align**, a comprehensive probability-alignment framework that jointly aligns marginal, conditional, and label distributions through Euclidean Alignment, Conditional Direction Matching, and label standardization. (*IEEE TBME, under review*)
- As equal-first author, co-developed **BFT**, which combines knowledge-guided EEG augmentation, Monte Carlo-dropout-inspired feature masking, and a source-trained learning-to-rank module to weight transformed predictions, suppress test-time uncertainty, and improve transfer without backpropagation. (*IEEE JBHI, accepted, 2026*)

Brain-to-Image Retrieval

Dec 2025 – Present

- Identified systematic cross-subject coordinate differences in EEG-to-image retrieval: subjects preserve similar concept relationships but express them along different directions. Proposed **SCORE** to select reliable EEG-image landmarks and estimate an orthogonal reorientation for label-free coordinate recovery.
- Identified a non-transitive alignment pattern in few-repetition retrieval: the low-repetition neural query and image representation each align with the high-repetition center, but not with each other. Proposed **NEAR** to use this center as an anchor, denoising queries and predicting pseudo-anchors for gallery images across EEG, MEG, and fMRI.

Selected Projects and Competitions

SSVEP-Controlled Robotic Arm with Adaptive Step Size

Sep 2025 – Dec 2025

- Built an SSVEP acquisition and decoding system for fast, precise robotic-arm control, using position-aware adaptive step sizing to balance movement speed and endpoint precision; won the **national championship** in the Brain-Controlled Robotic Arm track of the 2025 China Brain-Computer Interface Competition. The system was later deployed at Wuhan Optics Valley Biotech City; the competition was covered by CCTV and *Global Times*.

Porcine ECoG-Based Continuous Joint-Motion Decoding

2024

- Developed a porcine ECoG-based continuous joint-motion decoding algorithm with a segment-slicing augmentation method tailored to recurring joint-motion patterns during running; won **First Prize** in the minimally invasive BCI track at the 2024 World Robot Contest Championship.

Industry Experience

Alibaba Cloud Intelligence Group

Apr 2026 – Present

Research Intern

- Developed a foundation embedding model specialized for Elastic Compute Service (ECS) logs by designing a domain-specific tokenizer and contrastive pretraining objectives, then pairing it with a downstream classifier that integrates temporal attention. On production server-log data, the system improved F1 from **0.4383** to **0.5306** (+21.1% relative).
- Contributed to Alibaba Cloud's internal **AutoResearch** platform. Given a problem definition, a code branch, and evaluation metrics, the platform drives an agent to iteratively modify the solution, run evaluations, and improve task performance.

Dongfeng Motor Group — Multimodal Intelligent Cockpit Evaluation

Jun 2025 – Jun 2026

Industry-funded project — Project Lead

- Developed a multimodal intelligent-cockpit evaluation framework integrating EEG, images, audio, and text to assess users' subjective experience of in-vehicle functions.

Selected Awards and Honors

- National Champion**, China Brain-Computer Interface Competition, Brain-Controlled Robotic Arm track, 2025.
- First Prize**, Porcine ECoG-Based Continuous Joint-Motion Decoding, World Robot Contest Championship, 2024.
- First Prize**, National Undergraduate Intelligent Vehicle Competition, Multi-Vehicle Formation track, 2022.
- First-Class Graduate Academic Scholarship, HUST, 2024–25 and 2025–26; Yunnan Scholarship, 2024–25; Huawei Scholarship, 2021–22.
- Outstanding Graduate of HUST, 2024.